

CURITIBA, Brazil

WMO: 838400

Lat: 25.528S

Lon: 49.176W

Elev: 2988

StdP: 13.18

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.6	41.4	30.0	26.8	49.0	34.9	33.2	50.3	22.1	67.5	19.3	68.5	3.9	120	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.5	87.7	68.2	85.8	68.5	83.9	68.4	73.4	80.5	72.2	79.4	71.3	78.4	10.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	130.9	75.8	70.2	124.2	74.3	69.6	121.6	73.7	39.2	80.2	38.1	79.2	37.2	78.8	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.5	16.9	14.9	DB	31.1	91.4	3.3	1.2	28.7	92.3	26.7	93.0	24.9	93.7	22.4	94.5
			WB	30.1	75.7	3.5	1.7	27.6	76.9	25.6	77.9	23.6	78.9	21.1	80.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	71.4	72.0	69.9	66.6	60.5	58.4	57.9	59.9	61.8	64.7	66.7	70.0
	DBStd	7.28	4.10	4.11	4.34	5.08	5.11	5.68	6.10	6.42	6.64	5.61	5.33	4.76
	HDD50	30	0	0	0	0	1	7	13	7	2	0	0	0
	HDD65	1094	5	4	9	41	156	203	226	177	139	76	44	14
	CDD50	5490	664	616	618	499	326	261	258	314	356	457	500	621
	CDD65	1078	203	200	162	89	16	6	6	19	44	68	95	170
	CDH74	6245	990	1014	782	427	103	60	110	362	464	465	536	932
(13)	CDH80	1527	248	266	186	72	8	1	3	64	157	137	135	250
(14)	Wind	WSAvg	6.9	7.2	6.9	6.6	6.2	6.0	6.2	6.6	7.4	7.9	8.1	7.6
(15) Precipitation	PrecAvg	65.1	8.9	7.4	6.2	3.7	3.9	4.3	4.8	3.1	5.2	5.7	5.3	6.5
	PrecMax	87.7	19.3	13.0	12.2	8.8	10.9	10.4	11.1	9.9	15.2	12.0	8.9	12.0
	PrecMin	47.4	2.7	2.8	1.9	0.6	0.4	1.5	0.3	0.3	0.6	2.5	1.1	1.8
	PrecStd	9.6	4.1	2.4	2.3	2.2	2.4	2.0	2.5	2.4	3.7	2.0	1.9	2.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	89.2	88.0	85.3	80.7	78.8	80.3	84.6	89.3	88.0	87.8	88.5
		MCWB	70.0	69.6	69.5	67.5	65.3	62.9	60.5	62.9	64.1	67.3	67.6	68.7
	2%	DB	85.7	85.9	84.5	82.3	77.0	75.5	77.1	82.0	84.6	84.2	84.1	85.8
		MCWB	70.6	70.4	69.8	67.4	64.0	61.9	61.1	61.9	64.0	66.8	67.3	68.4
	5%	DB	82.6	83.8	82.0	79.0	73.4	72.8	73.8	78.5	80.2	79.2	80.5	82.7
		MCWB	70.0	70.4	69.3	66.5	62.9	61.3	60.3	60.8	63.0	66.0	66.7	68.7
	10%	DB	80.2	80.6	78.7	75.5	69.7	69.3	70.1	73.8	73.4	74.8	76.7	79.4
		MCWB	69.7	70.0	68.2	65.9	61.5	60.0	59.1	60.2	62.9	65.0	65.8	68.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.7	75.3	74.0	72.0	68.6	66.2	65.7	67.2	68.8	71.2	72.0	73.7
		MCDB	81.6	80.9	81.2	78.8	76.1	73.1	73.9	76.1	80.2	81.4	80.7	81.3
	2%	WB	72.9	73.4	72.3	70.1	66.1	64.2	63.4	64.4	66.6	69.2	70.4	71.8
		MCDB	80.1	79.7	79.4	76.4	71.9	70.8	71.8	75.8	77.1	78.5	78.1	80.0
	5%	WB	71.7	72.0	71.1	68.8	64.6	62.8	61.8	62.7	64.9	67.6	68.9	70.6
		MCDB	78.8	78.6	77.7	74.6	70.2	69.4	69.7	73.4	74.2	76.1	75.7	78.0
	10%	WB	70.6	71.0	69.8	67.5	63.1	61.3	60.2	61.1	63.3	65.9	67.5	69.4
		MCDB	76.9	77.3	75.6	72.9	68.3	67.1	67.2	70.3	71.1	72.2	73.5	75.8
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	16.5	16.2	17.0	16.7	18.1	19.5	21.4	18.1	16.9	17.1	17.3
		MCDBR	20.5	20.7	20.8	21.1	21.8	23.9	26.6	30.3	29.0	25.1	23.1	22.3
		MCWBR	8.2	8.0	8.4	8.8	10.3	11.7	12.4	12.6	10.6	10.5	9.7	8.9
	5% WB	MCDBR	18.3	18.3	18.5	17.6	18.1	19.4	21.5	25.0	23.2	21.7	19.6	19.2
		MCWBR	8.3	8.1	8.5	8.2	9.8	10.7	11.5	11.9	10.5	10.3	9.3	8.6
(40) Clear-Sky Solar Irradiance	taub		0.376	0.368	0.360	0.344	0.333	0.335	0.341	0.397	0.481	0.417	0.384	0.379
	taud		2.488	2.507	2.523	2.539	2.537	2.521	2.487	2.291	2.037	2.297	2.403	2.450
	Ebn at Noon		307	305	298	288	278	270	272	266	257	286	302	306
	Edh at Noon		37	36	33	30	27	27	29	38	53	43	40	39
(44) All-Sky Solar Radiation	RadAvg		1699	1672	1473	1249	967	869	947	1165	1249	1373	1619	1704
(45)	RadStd		175	134	175	127	110	87	116	98	180	184	162	162

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.92	-1.06	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-0.92	-1.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page